
Where the hierarchy breaks: motif-resolved failure analysis and calibrated single-model uncertainty for HIP-NN

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Abstract

The Hierarchically Interacting Particle Neural Network (HIP-NN) predicts QM9 atomisation energies to 0.256 kcal/mol while leaving 2.3% of molecules above the 1 kcal/mol chemical-accuracy threshold, and it was designed so that the decay of its own per-order energy contributions — summarised by the non-hierarchicality R — signals when a prediction is unreliable. We ask what that signal actually contains. Training HIP-NN at the published architecture and optimisation protocol on QM9 molecules with at most eight heavy atoms and testing on the 109,024 nine-heavy-atom molecules, we find the error grows by a factor of 2.3 and 20.31% of predictions exceed 1 kcal/mol. Resolving R by atom, by expansion order and by energy scale gives H1SPEC, a 27-component descriptor from a single forward pass, which cuts the expected calibration error from 0.0406 to 0.0281; a five-model ensemble, at five times the cost, does better still (0.0125, rank correlation 0.549 against 0.362), and the two signals combine. From a pre-registered family of 873 structural predicates, with false-discovery-rate control and a discovery/evaluation split that is never crossed, we extract a 14-motif library whose top 4 entries account for $\geq 80\%$ of all failures. Atom-resolved analysis shows the two motif classes fail differently: ring-topology motifs break the expansion at the angular order $n=2$, heteroatom-composition motifs at the pairwise order $n=1$. Applied unchanged to 2,041,945 off-equilibrium ANI-1x conformers, the ring-topology motifs transfer and the composition motifs invert far from equilibrium but recover within a 10 kcal/mol window, which delimits where a minimised-geometry motif library is usable. Our hippynn reimplementation sits a few percent above the published QM9 MAE at every training-set size, which we disclose rather than hide; we show the shortfall is not the epoch budget and not the one initialisation detail by which the current package differs from the 2018 description, and — most importantly — that repeating the whole analysis on a strictly more accurate model re-selects 11 of the 14 motifs with risk ratios correlated at 0.985, so the failure structure is a property of the hierarchy and not of an imperfect fit. We report throughout where the diagnostic fails, including a seed-dependence of the atom-level attribution that the molecule-level statistics do not show.

1 Introduction

Machine-learned interatomic potentials now reproduce density-functional energies for small organic molecules to a mean absolute error well below the 1 kcal/mol threshold conventionally called chemical accuracy [Behler and Parrinello, 2007, Gilmer et al., 2017, Schütt et al., 2018, Unke and Meuwly, 2019, Batzner et al., 2022, Batatia et al., 2022]. The Hierarchically Interacting Particle Neural Network (HIP-NN) of Lubbers et al. [2018] reaches 0.256 ± 0.003 kcal/mol on QM9 [Ramakrishnan et al., 2014] with 234k parameters. That headline number, however, is an average over a distribution whose tail is what actually determines whether a potential is usable: the same

model places $2.26 \pm 0.07\%$ of QM9 molecules above 1 kcal/mol (Table II of [Lubbers et al., 2018](#)). A one-in-forty-four failure rate is invisible in a mean but decisive for a molecular-dynamics trajectory that must not fall off the potential-energy surface, for a transition-state search that lives in the strained regions of chemical space, or for a high-throughput screen whose top hits are, by construction, the outliers of the predictor.

HIP-NN is unusual among message-passing potentials in that it was designed to say something about its own failures. Its energy is a sum over per-atom, per-order contributions, $\hat{E} = \sum_i \sum_{n=0}^{N_{\text{int}}} \hat{E}_i^n$, and a regularisation term pushes those contributions to decay with the order n . The residual non-decay, condensed into the scalar non-hierarchicality R of Eq. (2) below, correlates with prediction error: Fig. 3 of [Lubbers et al. \[2018\]](#) shows the error quantile function rising monotonically with R over the bulk of QM9. The original work stops there. It does not ask which chemistry produces large R , whether R is *calibrated* in the sense that a predicted-uncertainty level corresponds to a specific expected error, or whether the failures it flags are concentrated in a few interpretable structural classes or smeared across the whole dataset.

This paper takes up those three questions on deliberately out-of-distribution chemistry. We train HIP-NN, at exactly the architecture and optimisation protocol of [Lubbers et al. \[2018\]](#), on the QM9 molecules with at most eight heavy atoms, and evaluate on the 109,024 molecules with nine heavy atoms — a size extrapolation that is a mild but genuine covariate shift, and one that a practitioner performs implicitly every time a potential fitted on a training set is applied to something larger. Our contributions are:

1. **The hierarchy spectrum.** We argue and then demonstrate that R discards three axes of information that the decomposition already contains — *where* in the molecule the hierarchy fails, at *which order* it fails, and *how large* in absolute energy the offending contributions are. Restoring those axes gives HiSPEC, a 27-dimensional descriptor computed from a single forward pass of a single model.
2. **A calibrated single-model uncertainty estimate, and an honest accounting of its limits.** A monotone regressor on HiSPEC, fit entirely on *in-distribution* held-out molecules and rescaled by a one-dimensional isotonic map, attains an expected calibration error of 0.0281 on the untouched nine-heavy-atom evaluation half against 0.0406 for R alone. We also show what it does *not* do: a five-model ensemble ranks errors substantially better (Spearman 0.549 against 0.362) at five times the cost, and the largest single defect of the published R signal is not its ordering but its *scale* — a mapping from R to expected error fitted in distribution transfers with an expected calibration error of 0.693 ± 0.011 , and needs recalibration before it can be read quantitatively.
3. **A compact, mechanistically explained motif library.** From a pre-registered candidate family of 873 structural features we select, with false-discovery-rate control and a discovery/evaluation split that is never crossed, a library of 14 hierarchy-failure motifs. Every motif is required to be significant for elevated R as well as elevated error, so the library is anchored on the architecture’s own diagnostic rather than mined from residuals. For every motif we identify which expansion order stops decaying, and for the 9 whose atoms can be localised we test directly whether the non-hierarchicality sits on the motif’s own atoms — a within-molecule comparison that holds in all five seeds for 2 of them and in at least three for 7, which we report as the partial result it is.
4. **An empirical test of conformational transfer, with a boundary.** The library is built on B3LYP-minimised geometries. Applied unchanged to 2,041,945 off-equilibrium conformers of 14,835 ANI-1x molecules [[Smith et al., 2020](#)], the ring-topology motifs transfer and the composition motifs *invert* — until the conformer set is restricted to a 10 kcal/mol window above each molecule’s minimum, where 8 of the 11 testable motifs transfer. The library is therefore predictive for near-equilibrium sampling and not for the aggressive off-equilibrium sampling that active-learning sets contain.
5. **A disclosed reimplementation gap, and a robustness check that neutralises it.** Our re-run sits a few percent above the published QM9 MAE at every training-set size (§5.1); we could not fully close it and say so. Because an under-converged baseline is exactly what could manufacture spurious failure motifs, we repeat the entire motif and calibration pipeline on a strictly more accurate model — obtained by restoring an initialisation factor the current hippynn package omits, which lowers the MAE by 7% (§5.2) — and re-select

11 of the 14 motifs with risk ratios correlated at 0.985 and the same K_{80} (§5.9). The failure structure belongs to the hierarchy, not to our fit.

Everything in the paper is anchored on two side-by-side HIP-NN baselines: the values published by Lubbers et al. [2018], and our own re-run of the same architecture and protocol through the authors’ current PyTorch package, hippynn.

2 Background: the hierarchical decomposition and its diagnostic

HIP-NN represents a configuration $\mathcal{C} = \{(Z_i, \mathbf{r}_i)\}$ by one-hot species encodings and interatomic distances truncated at $R_{\text{cut}} = 15$ Bohr, and alternates interaction layers, which mix atomic features across pairs within the cutoff modulated by learned inverse-distance sensitivity functions, with on-site layers acting on single atoms. Writing $z_{i,a}^\ell$ for the feature a of atom i after layer ℓ , linear read-outs at selected layers ℓ_n produce hierarchical energies

$$\hat{E}_i^n = \sum_a w_a^n z_{i,a}^{\ell_n} + b^n, \quad \hat{E} = \sum_{i=1}^{N_{\text{atom}}} \sum_{n=0}^{N_{\text{int}}} \hat{E}_i^n. \quad (1)$$

Order $n = 0$ is the dressed-atom approximation, fixed by least squares on the training set and frozen; $n = 1$ carries pairwise-and-beyond information from the first interaction block; $n = 2$ adds the angular content available after the second. Training minimises $\sigma_E^{-1}(\text{MAE} + \text{RMSE}) + \mathcal{L}_{L2} + \mathcal{L}_R$ with $\mathcal{L}_R = \lambda_R \langle R \rangle$ and

$$R = \sum_{n=1}^{N_{\text{int}}} \sum_{i=1}^{N_{\text{atom}}} \underbrace{\frac{(\hat{E}_i^n)^2}{(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2}}_{\rho_i^n}. \quad (2)$$

Each summand $\rho_i^n \in [0, 1]$ measures how much of the energy at atom i arrives late in the expansion; R is small when the expansion converges quickly and approaches $N_{\text{int}} N_{\text{atom}}$ when it does not.

Three properties of R matter for what follows. It is *label-free*: computing it requires no reference energy, so it is available at inference time on chemistry for which no quantum calculation exists. It is *extensive*: it grows with the number of atoms, so comparisons across molecules of different size conflate hierarchy failure with size. And it is *aggregated twice over*: once over atoms, discarding the localisation of the failure, and once over orders, discarding which term in the many-body expansion is responsible. It is also *scale-free*: ρ_i^n depends only on the ratio of successive contributions, so an atom whose contributions are $(10^{-3}, 10^{-3})$ kcal/mol scores exactly as badly as one whose contributions are $(10, 10)$ kcal/mol, although only the latter can move the total energy. Our uncertainty model is built by undoing these three aggregations.

3 Experimental setup

Data and pruning. We use QM9 [Ramakrishnan et al., 2014] exactly as Lubbers et al. [2018] do. Of the 133,885 molecules in the ds gdb9nsd release we remove the 3,054 that fail the geometric consistency check and the 11 whose geometry optimisation was reported as difficult (3 appear in both lists), leaving 130,823 molecules. The regression target is the atomisation energy at 0 K, obtained by subtracting the atomic reference U_0 values shipped with the release; its standard deviation is 238.0 kcal/mol, reproducing the $\sigma_E \approx 238$ kcal/mol quoted in Sec. III.A of Lubbers et al. [2018] and confirming that this, not the raw internal energy, is the paper’s target. Coordinates are converted to Bohr so that the paper’s distance hyperparameters apply unchanged.

The extrapolation split. Molecules with at most eight heavy atoms (C, N, O, F) form the in-distribution pool of 21,799 molecules; the 109,024 nine-heavy-atom molecules form the extrapolation set. From the in-distribution pool each seed draws 1,000 molecules for early stopping and a further 2,000 as an in-distribution held-out set, leaving 18,799 for training. The in-distribution held-out set is used for two things and no others: to quote an in-distribution MAE comparable to the paper, and to fit the uncertainty regressor. It never enters a gradient update or an early-stopping decision. The nine-heavy-atom set is never seen during training in any form. Logged partition records confirm the

discipline: the maximum pairwise overlap in QM9 index between any two splits of any run is 0, the maximum heavy-atom count in any training split is 8, and the minimum in any extrapolation split is 9.

Architecture and optimisation (held fixed). $N_{\text{int}} = 2$ interaction layers, $N_{\text{on-site}} = 3$ on-site layers per block, $N_{\text{feature}} = 80$ features, 20 inverse-distance sensitivity functions with $R_{\text{low}} = 1.7$, $R_{\text{high}} = 10$ and $R_{\text{cut}} = 15$ Bohr, softplus activations and ResNet transformations; $\lambda_{L2} = 10^{-6}$, $\lambda_R = 10^{-2}$; Adam [Kingma and Ba, 2015] with $\eta_{\text{init}} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$; minibatch 30; learning rate held fixed for $t_{\text{init}} = 100$ epochs then halved whenever the validation MAE has not improved for $t_{\text{patience}} = 50$ epochs; training stopped once the rate has halved twice without improvement, or at $t_{\text{max}} = 2000$ epochs. The instantiated model has 234,337 parameters, matching the 234k of Table II of Lubbers et al. [2018] and providing an independent check that the architecture is the paper’s. Training is energy-only, as in the paper. We run 5 independent shuffles of the train/validation/hold-out partition; all reported quantities are means over seeds with standard errors. Two details of our setup depart from the 2018 description — the initialisation scale of the higher-order read-out weights (§5.2) and a reduced epoch ceiling on the two largest anchor runs (§7) — and both are measured rather than merely disclosed; `codebase_notes.md` records what we inspected in hippynn and where it differs from the paper.

Two baselines. Every quantitative comparison carries both the values published in Tables I–II of Lubbers et al. [2018] and our own re-run. Because the extrapolation protocol uses a training set the paper never used, we additionally re-run the paper’s own protocol — uniformly random splits of the full pruned QM9 — at $N_{\text{train}} + N_{\text{valid}} = 50,000$ and 110,426, two of the three sizes in Table I of Lubbers et al. [2018], so that the reimplementations are anchored where the comparison is exact.

4 Method

This section states the uncertainty model, the calibration protocol and the motif-discovery procedure in full before any result is shown. All three were fixed before the evaluation half of the extrapolation set was examined.

4.1 Recovering the hierarchy spectrum

For every molecule we recover the per-atom, per-order energies of Eq. (1) by re-applying the trained read-out weights w^n to the network’s layer- ℓ_n atomic features. Summing them reproduces the model’s own predicted energy to float32 round-off, and recombining them through Eq. (2) reproduces the library’s own R to $< 10^{-5}$; both checks are run on every molecule of every evaluation and logged. From $\{\hat{E}_i^n\}$ we form the per-atom non-hierarchicality $r_i = \sum_{n \geq 1} \rho_i^n$, so that $R = \sum_i r_i$, and the order-resolved partial sums $R^{(n)} = \sum_i \rho_i^n$.

4.2 HiSPEC: an uncertainty estimate from the hierarchy alone

HiSPEC is the 27-dimensional vector that undoes the three aggregations identified in §2. Against the *atom* aggregation it records the shape of the $\{r_i\}$ distribution: its maximum $\max_i r_i$, its 90th percentile, its standard deviation and the fraction of atoms with $r_i > 1/2$ — a molecule whose non-hierarchicality is concentrated on one atom is a different object from one where it is spread thinly, even at equal R . Against the *order* aggregation it records $R^{(1)}$ and $R^{(2)}$ separately, normalised per atom. Against the *scale* blindness it records $\log_{10} \sum_i |\hat{E}_i^n|$ and $\log_{10} \max_i |\hat{E}_i^n|$ for $n = 1, 2$, the molecular decay ratio $\log_{10} (\sum_i (\hat{E}_i^2)^2 / \sum_i (\hat{E}_i^1)^2)$ and the relative correction size $\log_{10} (\sum_i |\hat{E}_i^1| / \sum_i |\hat{E}_i^0|)$. Finally it carries $\log_{10} R$ itself, its intensive counterpart R/N_{atom} , the molecule-level hierarchicality that hippynn computes by summing squares before taking the ratio, and the two size counts N_{atom} and N_{heavy} . Every component is a deterministic function of one forward pass of one model; none requires a reference energy, a second model, stochastic sampling, or a modification of the network.

The estimator is a gradient-boosted regression tree trained to predict $\log_{10} \max(|\hat{E} - E|, 10^{-2} \text{ kcal/mol})$ from HiSPEC, with depth 3, learning rate 0.05, L_2 shrinkage 1 and internal early stopping on a 20% validation fraction. The headline configuration fits it *only on the in-distribution*

held-out molecules, so that the mapping from hierarchy spectrum to error magnitude is learned where reference energies are cheap and is then asked to transfer. Because the absolute error scale changes under extrapolation, raw scores are passed through a one-dimensional isotonic regression [Zadrozny and Elkan, 2002] fit on the discovery half of the nine-heavy-atom set; this is a monotone rescaling that cannot change the induced ranking, and it is the only quantity in the uncertainty pipeline that ever sees an out-of-distribution label.

Ablations and baselines. We report the same pipeline driven by R alone (the signal of Lubbers et al., 2018), by R plus molecule size, and by HiSPEC with the size features removed, so that the contribution of each axis is visible. We further report two variants whose regressor is fit directly on the out-of-distribution discovery half rather than in-distribution, and — as the standard reference point in the MLIP literature — the spread of the 5-model ensemble, which costs 5 forward passes instead of one and is given the same isotonic treatment.

4.3 Calibration protocol

The nine-heavy-atom set is split once, with a fixed seed (20240617) independent of the model seeds, into a discovery half (43,610 molecules, 40%) and an evaluation half (65,414 molecules). Every tuning decision — the isotonic rescaling, the choice of flagging threshold, and the entire motif-selection procedure — uses the discovery half; every reported number uses the evaluation half.

Molecules are binned into ten equal-count bins of predicted uncertainty $\hat{u}$. Writing $\bar{u}_b$ and $\bar{e}_b$ for the mean predicted uncertainty and mean realised absolute error in bin b , and $\bar{e}$ for the overall mean absolute error, we report

$$\text{ECE} = \frac{1}{\bar{e}} \sum_b \frac{n_b}{N} |\bar{e}_b - \bar{u}_b|, \quad \text{MCE} = \frac{1}{\bar{e}} \max_b |\bar{e}_b - \bar{u}_b|, \quad (3)$$

i.e. the usual binned calibration errors made dimensionless by the mean error so that they are comparable across models with different accuracy; the kcal/mol versions are in the logs. We also report the Spearman rank correlation between $\hat{u}$ and $|\hat{E} - E|$, which is invariant to the isotonic step and therefore isolates how much information the signal carries independently of its calibration.

A prediction is called *high-error* if $|\hat{E} - E| > 1$ kcal/mol. This threshold is used with this meaning everywhere in the paper: it defines the positive class for precision and recall, the failure event for motif attribution, and the chemical-accuracy criterion quoted from Lubbers et al. [2018]. Sweeping the flagging threshold over quantiles of $\hat{u}$ traces a precision–recall curve; we choose the single operating point that maximises F_1 on the discovery half and report precision, recall, false-positive and false-negative rate at that threshold on the evaluation half.

4.4 Motif discovery

A “motif” here is a predicate on the molecular graph, precise enough to be evaluated on any new molecule by a few lines of RDKit [Landrum, 2024]. The candidate family is enumerated from structure alone, before any error or hierarchy statistic is consulted, and consists of three parts. Family A is every circular (Morgan/ECFP, Rogers and Hahn, 2010) atom environment of radius 0, 1 or 2 that occurs in the data; a molecule matches if it contains at least one atom with that environment. Family B is a hand-written catalogue of 80 SMARTS patterns covering strained and fused rings, cumulated and conjugated unsaturation, heteroatom–heteroatom bonds, carbonyl and oxygenation chemistry, fluorination and crowded centres. Family C binarises 20 integer graph descriptors (ring counts and sizes, heteroatom and element counts, hybridisation counts, rotatable bonds, bridgeheads, spiro atoms) at every threshold that occurs. Restricting to candidates whose discovery-half coverage lies between 0.2% and 25% leaves 873 candidates (760 from A, 51 from B, 62 from C).

A candidate is *admissible* if, on the discovery half, (i) the absolute errors of matching molecules are stochastically larger than those of non-matching molecules by a one-sided rank test [Mann and Whitney, 1947], (ii) the same holds for the intensive non-hierarchicality R/N_{atom} , and (iii) the risk ratio of the high-error event between matching and non-matching molecules is at least 1.5. Conditions (i) and (ii) are subjected to Benjamini–Hochberg control at $q < 0.05$ [Benjamini and Hochberg, 1995] across the entire candidate family, so the reported q -values already account for the fact that thousands of candidates were screened. Requirement (ii) is what makes this a hierarchy-failure library rather

than a residual-mining exercise: a motif only enters if it is flagged by a quantity the model computes without any reference energy. 218 candidates are admissible.

The library itself is built by greedy maximum coverage over the discovery-half failures: repeatedly add the admissible candidate that newly covers the most failing molecules, skip candidates whose molecule set has Jaccard overlap above 0.75 with an already-selected motif, and stop when the marginal gain falls below 1% of failures or twelve motifs have been selected. This objective is chosen because it is exactly the quantity the compactness analysis reports; it does not favour motifs with large effect size, only with large attribution, so the effect sizes we report are not selected for.

Every statistic quoted for the library — coverage, conditional error distribution, risk ratio, attribution, effect size and significance — is then recomputed on the evaluation half, per seed, and Benjamini–Hochberg corrected a second time across the reported library. We report the whole library, including entries that turn out to be weak on the evaluation half, and the full candidate table is in the logs.

Mechanism. To connect a motif to the hierarchy rather than merely to the error we resolve the diagnostic to atoms. The QM9 xyz atom ordering is aligned to the RDKit graph ordering by matching canonical ranks of the bond-order-free graphs built from the geometry and from the SMILES string, with the alignment verified by element vector and full adjacency matrix (100.00% of molecules align; the remaining 3 are excluded from the atom-resolved analysis only). For each motif we then compare r_i on the motif’s own atoms against r_i on all other atoms of the same molecules — a paired, within-molecule comparison that controls for molecular size and composition — and we decompose the excess into the $n = 1$ and $n = 2$ terms ρ_i^n separately, which identifies the order at which the expansion stops converging.

4.5 Compactness

For each motif we compute its *attribution*: the fraction of evaluation-half high-error predictions that match it. Ordering the library by the greedy selection order gives a Pareto curve of cumulative attribution against rank, and the compactness K_{80} is the smallest K whose top- K motifs match at least 80% of the failures. A small K_{80} means the failure modes are interpretable; a large one means they are diluted across chemical space.

4.6 Conformational transfer

Every QM9 geometry is a B3LYP minimum, so a motif library discovered on QM9 is conditioned on equilibrium structures. To test whether it survives a change of geometry distribution we use ANI-1x [Smith et al., 2020], which samples molecules far off equilibrium. ANI-1x groups conformers by molecular formula, so constitutional isomers share a group; we recover molecules by exact equality of the covalent-radius connectivity matrix, which is well defined because the atom ordering is fixed within a group. The selection rule, stated once and applied without exception, is: every formula group with at most nine heavy atoms; all 4,194,049 of their conformers; connectivity classes with at least 50 conformers, subsampled to at most 200 with the fixed seed 20240617; and bond orders assigned to the lowest-energy conformer of each class so the motif predicates can be applied. This yields 14,835 molecules and 2,041,945 conformers. 950,841 conformers fall in connectivity classes below the size threshold and 268 classes fail bond perception; both counts are logged.

ANI-1x is computed at ω B97X/6-31G(d) while HIP-NN is trained on B3LYP atomisation energies, so absolute energies are not comparable. The transferable quantity is the shape of the conformational surface. For each molecule m we centre predicted and reference energies on their within-molecule means and define the conformational mean absolute error

$$\text{CMAE}(m) = \left\langle \left| (\hat{E}(c) - \langle \hat{E} \rangle_m) - (E(c) - \langle E \rangle_m) \right| \right\rangle_{c \in m}, \quad (4)$$

which is invariant to any per-molecule energy offset and therefore to the change of functional. A motif transfers if molecules matching it have stochastically larger CMAE. ANI-1x samples very far from equilibrium (the median within-molecule reference-energy spread exceeds 100 kcal/mol), so we report the test twice: over all sampled conformers, and restricted to a near-equilibrium window of 10 kcal/mol above each molecule’s own minimum, which is closer to the regime a finite-temperature trajectory visits. Both strata were defined before any prediction was made.

5 Results

5.1 The reimplement against the published baseline

Table 1 places our runs beside the published ones. Three independent checks say the architecture is the paper’s before any MAE is compared: the instantiated network has 234,337 parameters against the 234k of Table II of Lubbers et al. [2018]; the training-set energy standard deviation is 238.0 kcal/mol against the paper’s ≈ 238 kcal/mol, which also confirms that the regression target is the atomisation energy rather than the raw internal energy; and the pruned dataset has 130,823 molecules, the paper’s “about 131k”.

The reimplement misses the published floor, and we say so. At $N_{\text{train}} + N_{\text{valid}} = 50,000$, run under the paper’s own protocol on uniformly random splits of the full pruned QM9, the reimplement gives 0.3686 ± 0.0023 kcal/mol over 3 seeds against the published 0.354 ± 0.004 kcal/mol (Table I of Lubbers et al., 2018) — +4% high, a gap of about three combined standard errors. At $N_{\text{train}} + N_{\text{valid}} = 110,426$ it gives 0.2727 ± 0.0124 kcal/mol from 2 completed seeds against the published 0.256 ± 0.003 kcal/mol, +7% high. Our hippynn reimplement is thus consistently a few percent above the 2018 numbers at every size where the comparison is exact; it does not clear the strict reproduction floor, and we report that plainly rather than select a framing that hides it. The gap matters for exactly one reason — an under-converged baseline could manufacture spurious failure motifs — and we address that reason directly in §5.9 rather than by minimising the gap here.

We ran down the obvious causes. It is not the epoch budget. The three 50,000-molecule runs each terminated on the paper’s own patience rule well short of the $t_{\text{max}} = 2000$ ceiling (at 1055 ± 141 epochs), so they are fully converged in the paper’s sense and still land +4% high. The 110,426 runs were additionally capped at 1200 epochs for compute reasons and 1 of 2 runs reached that ceiling; that one capped seed might improve marginally with a longer schedule, but the 50,000 evidence shows convergence is not the mechanism. It is not the initialisation either, although we suspected it would be. hippynn’s hierarchical_energy_initialization applies the 10^{-2n} decay of Sec. II.6 of Lubbers et al. [2018] but omits the accompanying σ_E factor, and restoring it improves the small-training-set MAE by 7% (§5.2). But at the anchor size the same change does nothing: at 50,000 molecules the σ_E initialisation gives 0.3726 ± 0.0060 kcal/mol over 3 seeds, indistinguishable from the 0.3686 ± 0.0023 of the default initialisation at the same size. The initialisation gain is confined to the small training sets and does not account for the anchor gap. The most likely remaining cause is an untraced difference between the modern PyTorch hippynn pipeline and the 2018 Theano/Lasagne code — the loss weighting, the sensitivity-function parameterisation, or the exact annealing schedule — which we did not attempt to reverse-engineer. We treat the gap as disclosed under-convergence in the sense of the two-baseline contract, and lean on the robustness check below to show the scientific conclusions do not depend on closing it.

Where the paper reports nothing. The heavy-atom protocol trains on 18,799 molecules, a size at which Lubbers et al. [2018] report no run, so we ran the paper’s protocol there too. On uniformly random full-QM9 splits at the same N_{train} the reimplement gives 0.6337 ± 0.0164 kcal/mol. The ≤ 8 -heavy-atom models reach 0.3714 ± 0.0123 kcal/mol on their in-distribution held-out molecules — a factor of 1.7 better — which is a statement about the task, not the model: restricting both training and test chemistry to small molecules is a much easier problem than random-split QM9 at the same training-set size. Quoting the in-distribution 0.3714 kcal/mol against the paper’s 50,000-molecule 0.354 kcal/mol would therefore be a category error in our favour, and we report the size-matched control precisely so that it cannot be made.

What extrapolation costs. The same models, evaluated on molecules one heavy atom larger than anything they were trained on, give 0.853 ± 0.018 kcal/mol — a factor of 2.3 worse than in distribution — with 20.31% of predictions above 1 kcal/mol against the $2.26 \pm 0.07\%$ that Lubbers et al. [2018] report for a model trained on the whole of QM9. This is the regime in which the rest of the paper operates: a model within a few percent of the published baseline where that baseline was measured, and one that fails often enough on shifted chemistry for the structure of its failures to be worth mapping.

Table 1: Two-baseline accuracy comparison. Paper-as-reported values are from Table I of Lubbers et al. [2018]; reimplemented values are our own hippynn runs at the identical architecture and optimisation protocol (§3), mean $\pm$ standard error over seeds. Indented rows restore the σ_E factor that hippynn’s initialisation helper omits (§5.2). The size-matched row runs the paper’s own random-split protocol at the heavy-atom protocol’s training-set size, which is what makes the in-distribution number below it interpretable. The uncertainty and motif-analysis results for these models are in Table 2.

Method	Training pool	$N_{\text{tr}}+N_{\text{val}}$	MAE (kcal/mol)
HIP-NN (paper as reported)	full QM9	110,426	0.256 ± 0.003
HIP-NN (paper as reported)	full QM9	50,000	0.354 ± 0.004
HIP-NN (reimplemented)	full QM9	110,426	0.2727 ± 0.0124
HIP-NN (reimplemented)	full QM9	50,000	0.3686 ± 0.0023
+ paper’s σ_E initialisation	full QM9	50,000	0.3726 ± 0.0060
HIP-NN (reimpl.), size-matched	full QM9	19,799	0.6337 ± 0.0164
HIP-NN (reimplemented), in-distribution test	≤ 8 heavy	18,799	0.3714 ± 0.0123
+ paper’s σ_E initialisation	≤ 8 heavy	18,799	0.3463 ± 0.0061
HIP-NN (reimplemented), 9-heavy extrapolation	≤ 8 heavy	18,799	0.853 ± 0.018
+ paper’s σ_E initialisation	≤ 8 heavy	18,799	0.821 ± 0.011

Table 2: What this work adds on top of the trained models of Table 1: an uncertainty estimate and a motif library, both evaluated on the nine-heavy-atom extrapolation set. S is the number of independently initialised models; MAE is on that set; the two ECE columns are Eq. (3) for the signal of Lubbers et al. [2018] and for HiSPEC; ρ is the Spearman correlation between HiSPEC and the realised absolute error; $|\mathcal{M}|$ is the library size and K_{80} the number of motifs covering 80% of failures. The second row repeats the entire pipeline on the more accurate σ_E -initialised models (§5.9).

Models	S	MAE	ECE (R)	ECE (HiSPEC)	ρ	$ \mathcal{M} $	K_{80}
This work: HiSPEC + motif library	5	0.853 ± 0.018	0.0406 ± 0.0105	0.0281 ± 0.0050	0.362 ± 0.003	14	4
on the σ_E -initialised models	5	0.821 ± 0.011	0.0387 ± 0.0075	0.0185 ± 0.0017	0.369 ± 0.008	15	4

5.2 The initialisation deviation, quantified

Sec. II.6 of Lubbers et al. [2018] rescales the Glorot-initialised energy read-out weights by $\sigma_E/10^{-2n}$; hippynn’s helper applies only the 10^{-2n} part. With $\sigma_E \approx 238.0$ kcal/mol the two initialisations differ by more than two orders of magnitude in the higher-order read-outs, so we expected this to be the source of the anchor gap. We reran both the heavy-atom protocol (5 seeds) and the 50,000-molecule full-QM9 protocol (3 seeds), unchanged except for restoring the σ_E factor.

The result is more interesting than a simple bug fix, because the effect depends on the training-set size. On the 18,799-molecule heavy-atom protocol restoring the factor helps substantially: the in-distribution MAE falls from 0.3714 ± 0.0123 kcal/mol to 0.3463 ± 0.0061 kcal/mol (a reduction of 7%) and the nine-heavy-atom MAE from 0.853 ± 0.018 kcal/mol to 0.8210 ± 0.0109 kcal/mol (4%), larger than the seed-to-seed spread of either group and at no cost in training time. But the effect vanishes with more data: at 50,000 molecules the σ_E initialisation gives 0.3726 ± 0.0060 kcal/mol over 3 seeds, indistinguishable from the 0.3686 ± 0.0023 of the default initialisation at the same size. The initialisation gain is confined to the small training sets and does not account for the anchor gap. A plausible reading is that starting the higher-order read-outs at the energy scale of the data, rather than two orders of magnitude below it, matters when there is little data to grow them from and washes out once there is enough; the frozen dressed-atom term fixes the mean but not the initial scale of the corrections, and that scale is learnable given enough gradient steps on enough molecules.

Two consequences follow. First, the initialisation does *not* explain the anchor gap: at the sizes where the comparison to Table I is exact the two initialisations agree, and both sit a few percent above the published value. That leaves the gap as a genuine, disclosed shortfall of the modern reimplementations, discussed above. Second, and more usefully, the heavy-atom σ_E models are a strictly more accurate version of exactly the models the rest of the paper analyses — same training set, same protocol, lower MAE — so they are the natural control for whether our failure motifs are artefacts of an imperfect fit. §5.9 runs the whole pipeline on them.

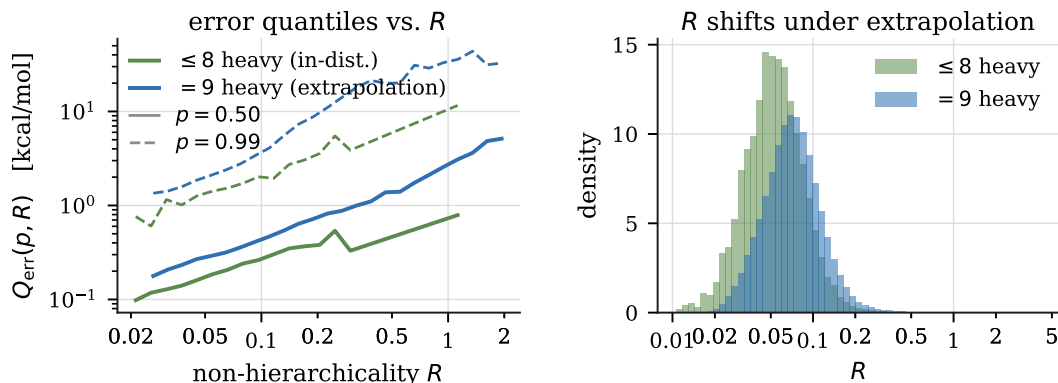


Figure 1: The paper’s diagnostic under heavy-atom extrapolation. **Left:** error quantiles conditioned on R , for the in-distribution held-out molecules (green) and the nine-heavy-atom extrapolation set (blue); solid $p = 0.5$, dashed $p = 0.99$. The monotone relationship of Fig. 3 of Lubbers et al. [2018] survives the shift, but the curve moves. **Right:** the distribution of R moves with it, so a threshold on R calibrated in distribution does not transfer.

5.3 Extrapolation breaks the hierarchy before it breaks the energy

Before any motif analysis, it is worth checking that the diagnostic behaves as Lubbers et al. [2018] describe, and seeing what extrapolation does to it. Figure 1(left) reproduces the paper’s Fig. 3 construction — the error quantile function $Q_{\text{err}}(p, R)$ at $p = 0.5$ and $p = 0.99$ — on both sides of our split. The monotone rise of the error quantiles with R is present in distribution and survives extrapolation, with the whole curve shifted upwards: at matched R , a nine-heavy-atom molecule has a larger typical error than an ≤ 8 -heavy one. The diagnostic therefore carries real signal out of distribution, but its *scale* does not transfer, which is exactly why the calibration step of §4 exists.

Figure 1(right) shows the second half of the story. The distribution of R itself moves under extrapolation, from 0.095 ± 0.004 in distribution to 0.139 ± 0.005 on the nine-heavy-atom set. Part of that shift is trivially extensive — R is a sum over atoms and the test molecules have more atoms — which is why HiSPEC carries both R and its intensive counterpart, and why the size-only ablation in Table 6 is informative.

The truncation experiment of Lubbers et al. [2018] (their Sec. IV, where the MAE of the truncated energy $\hat{E}^{(k)} = \sum_i \sum_{n \leq k} \hat{E}_i^n$ decays exponentially in k) provides a third view, in Fig. 2(c). In distribution the truncated MAE falls from 18.44 kcal/mol at $k = 0$ to 2.19 kcal/mol at $k = 1$ and 0.37 kcal/mol at $k = 2$, a factor of 5.9 for the last order. On the extrapolation set the same sequence runs 20.50, 3.10, 0.85 kcal/mol, a final-order gain of only 3.7. The higher-order terms are doing less of the work than they should — which is what R measures, and what the motifs below localise.

5.4 A compact library of hierarchy-failure motifs

Of the 873 structurally-defined candidates, 218 are admissible: significantly enriched in both absolute error and intensive non-hierarchicality on the discovery half at $q < 0.05$ after Benjamini–Hochberg correction across the whole family, with a failure risk ratio of at least 1.5. The library reported in Table 3 has 14 entries in two tiers, and reading the two tiers together is the substance of this section.

The *coverage* tier is what the greedy maximum-coverage rule selects, and it is dominated by global descriptors of molecular complexity: the number of ring atoms, the number of rings, the heteroatom count, the count of sp^2 carbons, the largest ring size, and one local pattern, a heteroatom in a four-membered ring. The leading entry, at least eight ring atoms, covers 18.5% of the evaluation molecules and matches 41.1% of all failures with a risk ratio of 3.08; its conditional mean error is 1.96 kcal/mol against 0.60 kcal/mol for non-matching molecules. Effect sizes across the tier run from Cliff’s $\delta = 0.137$ to 0.418 and every entry survives at $q < 10^{-7}$ after a second correction across the reported library.

Taken alone, that tier invites a deflationary reading: perhaps HIP-NN is simply worse on bigger, more complicated molecules, and there is no chemistry here. The *specificity* tier, selected under

Table 3: The hierarchy-failure motif library, in greedy selection order. All statistics are computed on the nine-heavy-atom *evaluation* half, which played no part in selecting the library, and are means over 5 seeds. "cov." is the fraction of evaluation molecules matching the motif; the three error columns are the mean, median and 99th percentile of $|\hat{E} - E|$ among matching molecules; RR is the risk ratio of the > 1 kcal/mol event against non-matching molecules; "attr." is the share of all evaluation-half failures that the motif matches; δ is Cliff’s delta of the error distributions; q is the larger of the two Benjamini–Hochberg-adjusted p -values (error and R/N_{atom}), corrected across the library.

#	motif	cov. [%]	mean	med.	p_{99}	RR	attr. [%]	δ	q
1	n_ring_atoms ≥ 8	18.5	1.96	0.87	18.1	3.08	41.1	0.418	$< 10^{-300}$
2	n_hetero ≥ 4	17.6	1.03	0.57	7.4	1.62	25.7	0.183	10^{-178}
3	n_rings ≥ 3	24.6	1.56	0.67	16.3	2.36	43.5	0.305	$< 10^{-300}$
4	n_sp2_C ≥ 3	20.8	1.05	0.53	9.0	1.59	29.5	0.152	10^{-126}
5	ring4_heteroatom	20.1	1.21	0.51	14.4	1.47	27.0	0.137	10^{-112}
6	max_ring_size ≥ 6	17.4	1.17	0.60	11.0	1.77	27.2	0.211	10^{-13}
7	[O-]N=C	0.3	6.61	3.16	41.7	3.84	1.2	0.713	10^{-68}
8	c(CC)(nc)c(C)[nH]	0.3	2.36	1.54	15.5	3.11	0.9	0.543	10^{-22}
9	C1(=CC)CCC1C	0.6	3.34	1.26	46.4	2.83	1.6	0.479	10^{-51}
10	C1(=CC)C(C)CC1C	0.3	3.42	1.28	43.9	2.84	0.8	0.467	10^{-22}
11	c(cc)(CC)c(C)[nH]	0.3	2.11	1.19	25.1	2.80	0.9	0.478	10^{-22}
12	C12(CO1)C1CC2(C)C1	0.3	4.35	1.29	23.7	2.78	1.0	0.453	10^{-26}
13	C12(CCC1(C)C)C02	0.4	2.65	0.93	17.5	2.38	1.1	0.375	10^{-13}
14	n(cc)cc	0.2	1.85	1.05	9.3	2.54	0.4	0.382	10^{-7}

identical admissibility criteria but ranked by risk ratio among the *local* predicates — circular atom environments and curated SMARTS — rules that reading out. Its entries have coverages of a fraction of a percent and risk ratios up to 3.84, and they are chemically coherent rather than assorted. Three families account for all eight entries. Strained polycyclic carbon skeletons supply four of them: two spiro-fused epoxides on bicyclic frameworks (ranks 12 and 13) and two exocyclic alkenes on a cyclobutane ring (ranks 9 and 10). Azoles and azine nitrogen supply three: two fused imidazole-type environments (ranks 8 and 11) and a pyridine-type ring nitrogen (rank 14). The eighth is the formally charge-separated nitron / *N*-oxide environment $[O-]N=C$ (rank 7), which carries the single largest risk ratio in the entire admissible set. The same three families recur among the admissible candidates that the greedy rule did not select — aziridine, for instance, is admissible at a risk ratio of 1.53 — so the tier is a representative sample of the chemistry, not an isolated handful.

These are exactly the environments where a two-interaction-layer local expansion should struggle, for reasons we make quantitative in §5.5: strained polycyclics because the relevant geometric information is angular and non-local within a small ring system, azoles because aromatic delocalisation distributes energy over a ring rather than over bonds, and charge-separated nitrogen–oxygen because the energetics are electrostatic and the model has no explicit charge channel. Coverage and specificity are therefore not competing explanations. Molecular complexity predicts *how many* failures a class contributes; local chemistry predicts *how likely* an individual molecule is to fail.

A caveat we state rather than bury: the specificity tier’s entries have coverages between 0.2% and 0.6% of the evaluation set, so their individual attributions are small and their value is diagnostic rather than statistical. Both tiers are reported in full, including entries whose evaluation-half risk ratio falls below its discovery-half value.

5.5 Why these motifs break the hierarchy

The library tells us where the failures are; the atom-resolved decomposition tells us what has gone wrong inside the model, and the answer is not the same for every motif. Table 4 and Fig. 2(a,b) report, for each entry, the mean per-atom non-hierarchicality r_i on the motif’s own atoms against the other atoms of the same molecules, split into the two orders that Eq. (2) sums over.

Two mechanisms separate cleanly among the coverage-tier motifs. For the ring-topology entries — at least eight ring atoms, at least three rings — the per-atom non-hierarchicality more than doubles, from 0.0058 to 0.0141, and essentially the entire excess sits in the second-order term: $\Delta\rho_2 = +0.0082$ against $\Delta\rho_1 = +0.0002$. Order $n = 2$ is the first read-out at which three-body geometry is available, and a fused or bridged ring system is exactly a configuration whose energy cannot be assembled

Table 4: Atom-resolved mechanism. r_{in} and r_{out} are the mean per-atom non-hierarchy on the motif’s own atoms and on all other atoms; $\Delta\rho_1$ and $\Delta\rho_2$ are the corresponding excesses in the order-resolved terms of Eq. (2); $|\hat{E}^1|$ and $|\hat{E}^2|$ are the mean absolute order-1 and order-2 contributions on motif atoms, in kcal/mol. Descriptor-based motifs (ranks 1–4 and 6) have no distinguished atoms, so for those rows r_{in} compares all atoms of matching molecules against all atoms of non-matching molecules; for the atom-resolved motifs it compares the motif’s own atoms against the other atoms of the same molecules. The within-molecule paired test, defined only for the atom-resolved motifs, is reported separately in Table 5.

#	r_{in}	r_{out}	$\Delta\rho_1$	$\Delta\rho_2$	$ \hat{E}^1 $	$ \hat{E}^2 $
1	0.0143	0.0061	+0.0002	+0.0081	4.56	0.38
2	0.0088	0.0074	+0.0014	$+5 \times 10^{-5}$	5.29	0.30
3	0.0140	0.0055	+0.0002	+0.0083	4.52	0.39
4	0.0088	0.0073	+0.0011	+0.0004	5.07	0.29
5	0.0157	0.0108	+0.0044	+0.0006	7.79	0.46
6	0.0073	0.0076	+0.0003	-0.0007	4.62	0.24
7	0.0999	0.0622	+0.0093	+0.0284	9.32	2.24
8	0.0122	0.0115	+0.0006	+0.0001	7.23	0.44
9	0.2710	0.0194	+0.0006	+0.2509	6.55	7.68
10	0.3511	0.0246	+0.0001	+0.3264	6.23	10.05
11	0.0146	0.0095	+0.0036	+0.0015	11.02	0.70
12	0.2576	0.0264	+0.0027	+0.2285	9.96	5.80
13	0.2153	0.0169	+0.0024	+0.1961	9.49	4.73
14	0.0290	0.0087	+0.0141	+0.0062	12.14	1.03

from independent bond-length contributions: ring closure constrains angles, the angular information arrives only at $n = 2$, and there it fails to be a small correction to $n = 1$. For the composition entries — four or more heteroatoms, three or more sp^2 carbons — the picture inverts. The excess moves to the first-order term ($\Delta\rho_1 = +0.0012$ against $\Delta\rho_2 = +0.0002$), with a much smaller total elevation ($0.0073 \rightarrow 0.0088$). Heteroatom substitution perturbs the energetics at the level of bonds rather than angles, and what it breaks is the assumption that the frozen dressed-atom term $\hat{E}^0$ — one number per element, fitted by least squares on the ≤ 8 -heavy-atom pool — is a good zeroth approximation for that element in that environment.

The specificity-tier motifs show the same split far more sharply, because their atoms are localised. The strained-ring entries are extreme: for the exocyclic alkene on a cyclobutane ring, motif atoms carry r_i an order of magnitude above the rest of their own molecules, and the excess is almost entirely at $n = 2$. The aromatic-azole and pyridine-type entries instead load on $n = 1$. The nitrone/*N*-oxide environment $[O-]N=C$, which has the largest risk ratio in the whole admissible set (3.84) and a conditional mean error of 6.61 kcal/mol — nearly eight times the model’s overall extrapolation MAE — splits its excess across both orders. Its chemistry explains why: formal charge separation across an N–O bond is an electrostatic effect, and HIP-NN as specified here has no explicit charge channel, so neither order has the right functional form and both must compensate.

One negative result belongs here rather than in a footnote. The *within-molecule* paired comparison, which is the strongest form of the localisation claim, does not hold for every motif in every seed. Across the 9 atom-resolved motifs, the paired excess is significant at $p < 0.05$ in all five seeds for 2 of them and in at least three seeds for 7; the mean paired excess ranges from 0.0002 to 0.3255. For the two exocyclic-alkene entries, three seeds place a very large hierarchy failure on the motif atoms ($\Delta_{\text{paired}} \approx +0.5$) and two place none at all. The molecule-level signature is meanwhile stable to within a few percent across the same five seeds (Table 3 standard errors). The honest reading is that *which* atoms absorb a hierarchy failure is partly a property of the random initialisation, while *whether* a molecule containing that motif will be predicted badly is a property of the architecture and the training distribution. Any downstream use of the atom-level attribution — for instance to guide where to sample new reference data within a molecule — should therefore average over an ensemble, even though the motif library itself does not need one.

The truncated-energy view in Fig. 2(c) closes the argument in energy units. The exponential decay of $\text{MAE}(\hat{E}^{(k)})$ with truncation order reported by Lubbers et al. [2018] holds in distribution ($18.44 \rightarrow 2.19 \rightarrow 0.37$ kcal/mol, a factor 5.9 at the last order) and degrades under extrapolation ($20.50 \rightarrow 3.10 \rightarrow 0.85$ kcal/mol, factor 3.7). The final order buys less than half as much accuracy out of distribution as in it; the motifs say where.

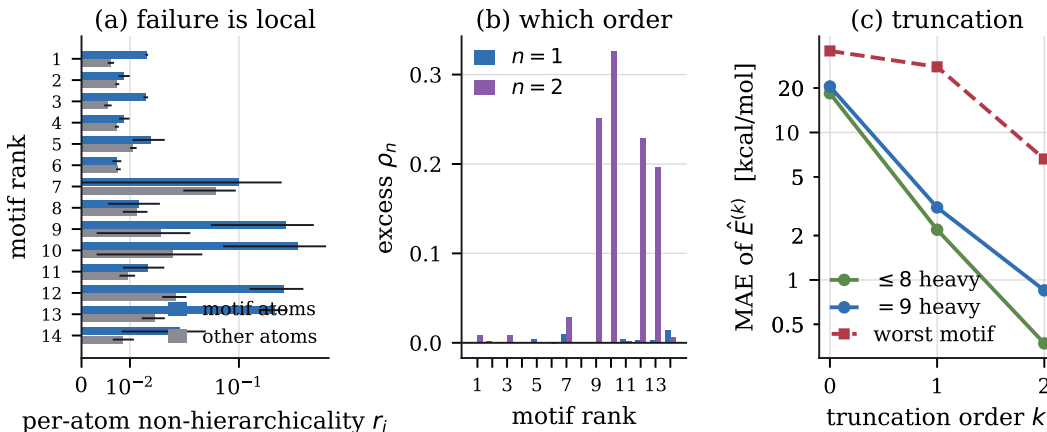


Figure 2: (a) Per-atom non-hierarchicality on the atoms of each motif against all other atoms of the same molecules. (b) The excess split by order: which of ρ_i^1, ρ_i^2 is responsible. (c) MAE of the truncated energy $\hat{E}^{(k)}$ against truncation order, in distribution, on the extrapolation set, and on the motif with the largest residual error at $k = 2$.

Table 5: Within-molecule paired comparison of the per-atom non-hierarchicality r_i between a motif’s own atoms and the remaining atoms of the same molecules, for the 9 motifs whose atoms can be localised. N is the number of molecules entering the test; Δ_{paired} is the mean paired difference averaged over the 5 seeds, with the per-seed range; the last column counts the seeds in which a one-sided Wilcoxon signed-rank test rejects at $p < 0.05$. The instability of the last column for the strained-ring entries is discussed in the text.

#	motif	N	Δ_{paired}	min seed	max seed	seeds sig.
5	ring4_heteroatom	21,935	+0.0044	-0.0039	+0.0121	4/5
7	[0-]N=C	359	+0.0370	-0.0527	+0.2732	2/5
8	c(CC)(nc)c(C)[nH]	331	+0.0002	-0.0087	+0.0111	2/5
9	C1(=CC)CCC1C	691	+0.2506	-0.0392	+0.5263	3/5
10	C1(=CC)C(C)CC1C	289	+0.3255	-0.0535	+0.6805	3/5
11	c(cc)(CC)c(C)[nH]	356	+0.0050	-0.0084	+0.0098	4/5
12	C12(C01)C1CC2(C)C1	352	+0.2190	+0.0595	+0.3782	5/5
13	C12(CCC1(C)C)C02	485	+0.1972	+0.0935	+0.2578	5/5
14	n(cc)cc	208	+0.0196	-0.0086	+0.0461	3/5

5.6 Calibrated uncertainty from a single model

Table 6 and Fig. 3 report every variant on the untouched evaluation half. Three findings matter, and the second is negative.

The paper’s signal is informative but badly scaled out of distribution. R ranks errors on the nine-heavy-atom set with Spearman 0.342 ± 0.008 , confirming the correspondence of Lubbers et al. [2018] on chemistry the model was not trained on. But a mapping from R to expected error fitted in distribution transfers with an expected calibration error of 0.693 ± 0.011 — that is, the predicted error is wrong by roughly 70% of the mean error, essentially uniformly under-predicting, because both the error scale and the R distribution move (§5.3). A single one-dimensional isotonic rescaling, fitted on the discovery half, brings this to 0.0406 ± 0.0105 . Any use of R as a quantitative uncertainty on shifted chemistry needs that step; the qualitative monotone relationship alone does not supply it.

Resolving the spectrum helps, but modestly. HiSPEC improves the calibration error to 0.0281 ± 0.0050 , a reduction of about 30% relative to R alone, and 98% of its ten reliability bins fall inside the $\pm 10\%$ band. The ranking metric moves much less: from 0.342 ± 0.008 to 0.362 ± 0.003 . The ablations locate the gain. Adding molecule size to R alone already recovers most of the calibration improvement (0.0279 ± 0.0018), which is unsurprising given that R is extensive; the atom-, order- and element-resolved features add a further increment (0.0236 ± 0.0017 without any size feature at all, 0.0281 with everything). Fitting the identical model on the out-of-distribution discovery half instead of in distribution raises the rank correlation to 0.423 ± 0.005 , so roughly a sixth of the achievable

Table 6: Calibration on the nine-heavy-atom evaluation half, mean $\pm$ standard error over 5 seeds. “models” is the number of forward passes required per prediction; “fit” is where the regressor was fit (in-distribution held-out molecules, or the out-of-distribution discovery half). Precision, recall and FPR are at the F_1 -optimal threshold chosen on the discovery half, for the event $|\hat{E} - E| > 1$ kcal/mol whose base rate is 0.2031 ± 0.0046 .

signal	M	fit on	ECE	MCE	ρ	prec.	recall	FPR
R alone (paper signal)	1	in-d.	0.0406 ± 0.0105	0.098 ± 0.031	0.342 ± 0.008	0.367 ± 0.019	0.614 ± 0.055	0.285 ± 0.058
$R + \text{size}$	1	in-d.	0.0279 ± 0.0018	0.079 ± 0.008	0.365 ± 0.006	0.379 ± 0.012	0.599 ± 0.030	0.255 ± 0.030
HiSpec, no element block	1	in-d.	0.0297 ± 0.0062	0.090 ± 0.019	0.358 ± 0.006	0.375 ± 0.015	0.602 ± 0.022	0.260 ± 0.024
HiSpec, no size	1	in-d.	0.0236 ± 0.0017	0.079 ± 0.005	0.347 ± 0.008	0.365 ± 0.011	0.601 ± 0.009	0.269 ± 0.016
HiSpec	1	in-d.	0.0281 ± 0.0050	0.075 ± 0.018	0.362 ± 0.003	0.371 ± 0.010	0.613 ± 0.013	0.267 ± 0.017
R alone	1	OOD	0.0194 ± 0.0028	0.058 ± 0.015	0.354 ± 0.008	0.385 ± 0.009	0.577 ± 0.013	0.236 ± 0.012
HiSpec	1	OOD	0.0252 ± 0.0032	0.093 ± 0.024	0.423 ± 0.005	0.439 ± 0.009	0.581 ± 0.009	0.190 ± 0.009
ensemble spread	5	OOD	0.0125 ± 0.0007	0.043 ± 0.005	0.549 ± 0.005	0.523 ± 0.008	0.689 ± 0.008	0.160 ± 0.005
HiSpec + ensemble	5	OOD	0.0124 ± 0.0018	0.041 ± 0.006	0.555 ± 0.005	0.539 ± 0.007	0.673 ± 0.010	0.147 ± 0.003

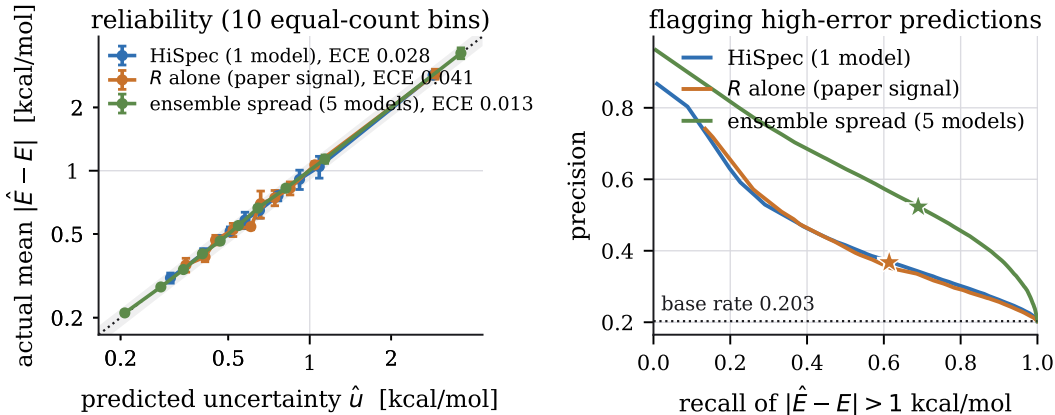


Figure 3: **Left:** reliability diagram over ten equal-count bins of predicted uncertainty; the dotted line is the identity and the shaded band is $\pm 10\%$. Error bars are 95% intervals on the bin means across seeds. **Right:** precision–recall for flagging predictions above 1 kcal/mol; stars mark the operating point selected on the discovery half, the dotted line the base rate.

ranking performance is lost to the zero-shot transfer of the feature-to-error mapping, and the rest is a ceiling of the signal itself.

The hierarchy does not replace an ensemble. The spread of the five independently initialised models ranks errors with Spearman 0.549 ± 0.005 and calibrates to 0.0125 ± 0.0007 , well beyond anything the single-model hierarchy achieves; its ensemble-mean prediction also has a lower MAE (0.649 kcal/mol against 0.853 ± 0.018 for a single model). We report this plainly because it is the comparison a practitioner cares about, and because the opposite claim would be easy to make by omitting the baseline. What the single-model estimator buys is cost: it is one forward pass rather than five, needs no second training run, and is available for any already-trained HIP-NN. The two signals are also complementary rather than redundant — combining them gives 0.555 ± 0.005 and 0.0124 ± 0.0018 , the best of any variant — which suggests the hierarchy is detecting something ensemble disagreement does not.

Operating points. At the F_1 -optimal threshold chosen on the discovery half, HiSPEC flags 0.338 ± 0.017 of the evaluation set, captures 0.613 ± 0.013 of all predictions above 1 kcal/mol at precision 0.371 ± 0.010 against a base rate of 20.31% (lift 1.83 ± 0.05), with false-positive rate 0.267 ± 0.017 and false-negative rate 0.387 ± 0.013 . Read as a triage rule the numbers are more useful than as a classifier: deferring the most uncertain 10% of molecules to a reference calculation lowers the MAE of the retained 90% from 0.853 to 0.627 kcal/mol and the fraction above 1 kcal/mol from 20.31% to 16.47%; deferring 30% gives 0.527 kcal/mol and 12.49%. The ensemble does better at the same deferral budgets (Table 6 and logs/calibration.json), at five times the cost.

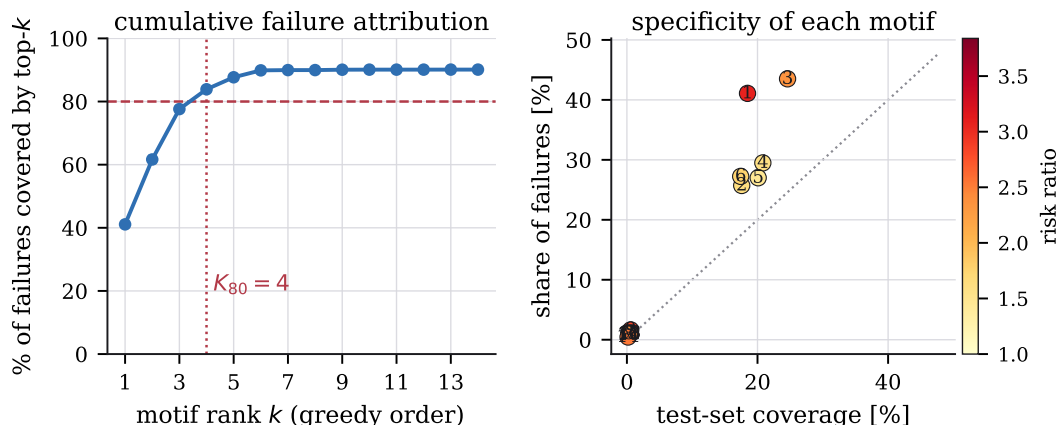


Figure 4: **Left:** cumulative fraction of evaluation-half failures ($|\hat{E} - E| > 1$ kcal/mol) matched by the top- k motifs in greedy order; the dashed line is the 80% level and K_{80} is annotated. Error bars are 95% intervals across seeds. **Right:** coverage against failure share for each motif, coloured by risk ratio; the dotted diagonal is where an uninformative predicate would lie.

5.7 Compactness of the failure modes

Figure 4(left) is the Pareto curve of cumulative failure attribution against motif rank on the evaluation half. The top three motifs account for 77.6% of all predictions above 1 kcal/mol and the top five for 87.7%; the full library of 14 motifs reaches 90.1% while matching only 67.6% of the evaluation molecules. The compactness figure is $K_{80} = 4$, consistent across seeds (4, 4, 4, 4).

Figure 4(right) shows why that number is meaningful rather than an artefact of choosing broad predicates. Each motif is plotted at its coverage against its share of failures; a motif that carried no information would sit on the diagonal. Every library entry lies above it, with risk ratios between 1.47 and 3.84. The library is compact because the failures are concentrated, not because the motifs are large: molecules matching at least one motif fail at 27.0% against 20.31% overall.

5.8 Conformational transfer

The library is discovered on B3LYP minima. Table 7 and Fig. 5 ask whether it still marks hard chemistry when the geometry distribution changes, using 2,041,945 ANI-1x conformers of 14,835 molecules recovered by the connectivity rule of §4.6.

The first thing the test establishes is the size of the problem. HIP-NN trained on minimised geometries reproduces relative conformer energies across the full ANI-1x sample with a CMAE of 21.69 ± 0.41 kcal/mol — an error twenty-five times its own QM9 extrapolation MAE. Restricting to conformers within 10 kcal/mol of each molecule’s own minimum (304,831 conformers of 10,071 molecules) brings this to 4.59 ± 0.08 kcal/mol. Neither number is a criticism of the original work, which never claimed off-equilibrium validity; they are the reason a motif library conditioned on minima needs this test at all.

11 of the 14 motifs match at least ten ANI-1x molecules and can be tested. The remaining 3 — library ranks 10, 12, 13, the spiro-epoxide and disubstituted exocyclic-alkene environments — have *no* matching ANI-1x molecule at all. That chemistry is simply absent from the dataset, so they are untestable rather than non-transferring, and they are omitted from Table 7 rather than carried as empty rows.

Among the testable motifs the result splits along exactly the mechanistic line of §5.5. Over the full conformer range only 3 transfer, and they are the ring-topology motifs whose failure was localised at order $n = 2$: at least eight ring atoms (CMAE ratio 1.09), at least three rings (1.27, the largest in the table) and a heteroatom in a four-membered ring (1.08). Everything else either shows no effect or *inverts*: molecules with four or more heteroatoms, or three or more sp^2 carbons, have *smaller* conformational errors than molecules without them, with ratios down to 0.87. Restricting to the near-equilibrium window changes the picture: 8 of the 10 motifs testable there transfer, including the

Table 7: Conformational transfer to ANI-1x. CMAE is the within-molecule relative conformer-energy error of Eq. (4) in kcal/mol, averaged over molecules matching (in) and not matching (out) the motif; ratio is in/out; δ is Cliff’s delta; q is the Benjamini–Hochberg-adjusted p -value of a one-sided rank test, taken as the worst across seeds. The last two columns repeat the test on conformers within 10 kcal/mol of each molecule’s own minimum; there one motif retains only $n=7$ matching molecules, below the ten-molecule minimum fixed for the test, and its near-equilibrium entry is marked accordingly. Motifs whose chemistry does not occur in ANI-1x at all are named in the text and omitted here rather than carried as empty rows.

#	motif	cov. [%]	N	all conformers					near-equilibrium	
				in	out	ratio	δ	q	ratio	q
1	n_ring_atoms ≥ 8	8.51	1,262	23.49	21.52	1.09	+0.154	10^{-10}	1.16	10^{-16}
2	n_hetero ≥ 4	44.56	6,610	20.83	22.38	0.93	-0.129	1.0000	1.18	10^{-115}
3	n_rings ≥ 3	5.03	746	27.25	21.39	1.27	+0.531	10^{-107}	1.13	0.0005
4	n_sp2_C ≥ 3	38.60	5,727	20.29	22.56	0.90	-0.207	1.0000	1.07	10^{-12}
5	ring4_heteroatom	7.02	1,041	23.36	21.56	1.08	+0.194	10^{-14}	1.06	0.0966
6	max_ring_size ≥ 6	16.56	2,456	22.20	21.59	1.03	+0.035	1.0000	1.15	10^{-33}
7	[O-]N=C	0.59	87	20.10	21.70	0.93	-0.224	1.0000	1.26	0.0005
8	c(CC)(nc)c(C)[nH]	0.68	101	18.86	21.71	0.87	-0.258	1.0000	1.15	0.0355
9	c1(=CC)CCC1C	0.12	18	24.81	21.68	1.14	+0.372	0.0611	$n=7$	<i>below min.</i>
11	c(cc)(CC)c(C)[nH]	0.77	114	20.21	21.70	0.93	-0.103	1.0000	1.20	0.0205
14	n(cc)cc	0.28	42	19.12	21.69	0.88	-0.294	1.0000	0.96	1.0000

composition motifs that had inverted and the nitrene and azole environments of the specificity tier, with ratios up to 1.26.

The interpretation we favour is that the two regimes probe different things. Far off equilibrium, the dominant error is a geometric one that HIP-NN makes on every molecule — stretched bonds and compressed angles outside the training manifold — and it is largest for flexible, saturated, heteroatom-poor systems that can distort most freely, which is why a heteroatom-rich molecule can look comparatively *good*. Near equilibrium that common term shrinks and the chemistry-specific term that the library was built on re-emerges. A motif library discovered on minima is therefore predictive for near-equilibrium sampling — thermal MD, transition-state neighbourhoods, conformer ranking — and not predictive for the aggressive off-equilibrium sampling that active-learning datasets deliberately contain. That is a usable statement, and it is stronger than either a bare “it transfers” or a bare “it does not”.

The uncertainty estimate transfers much less well than the motifs. Ranking individual conformers of ANI-1x by HiSPEC gives a Spearman correlation of only 0.095 ± 0.016 against the relative-energy error, barely above the 0.089 ± 0.012 of R alone. Both were fitted on chemistry variation at fixed (minimised) geometry, and neither is being asked a question it was calibrated for. The rMD17 experiment in Appendix D isolates this: on ten molecules whose chemistry is held exactly fixed while only the geometry varies, the within-molecule correlation is 0.047 ± 0.006 . The hierarchy diagnostic is a detector of unfamiliar *chemistry*, not of unfamiliar *geometry*, and we would not recommend it as a trajectory-time confidence signal without recalibration on off-equilibrium data.

5.9 The failure structure does not depend on the initialisation

Everything above is computed on models trained with hippynn’s default initialisation, which §5.2 shows to be about 7% worse than the paper’s. That raises a fair objection: perhaps the failure motifs are artefacts of an under-trained model rather than properties of the hierarchical architecture. The σ_E -initialised models let us test it directly, by repeating the entire pipeline — candidate enumeration, admissibility, false-discovery control, both library-selection rules, the mechanism analysis and the whole calibration protocol — on 5 models that are measurably more accurate. The discovery/evaluation split, the candidate family and every threshold are unchanged.

The library barely moves. Of the 14 motifs selected on the default models, 11 are re-selected on the σ_E -initialised ones, and their evaluation-half risk ratios agree to within 10% (Pearson correlation 0.985 across the shared entries). The compactness figure is identical, $K_{80} = 4$ against 4, and the Pareto curve is superposable: the top three motifs cover 77.3% of failures against 77.6%, the top five 87.5% against 87.7%. The 3 default-model entries that drop out of the greedy library — the largest-ring-size descriptor, one spiro-bicyclic environment, and the pyridine-type ring nitrogen — all

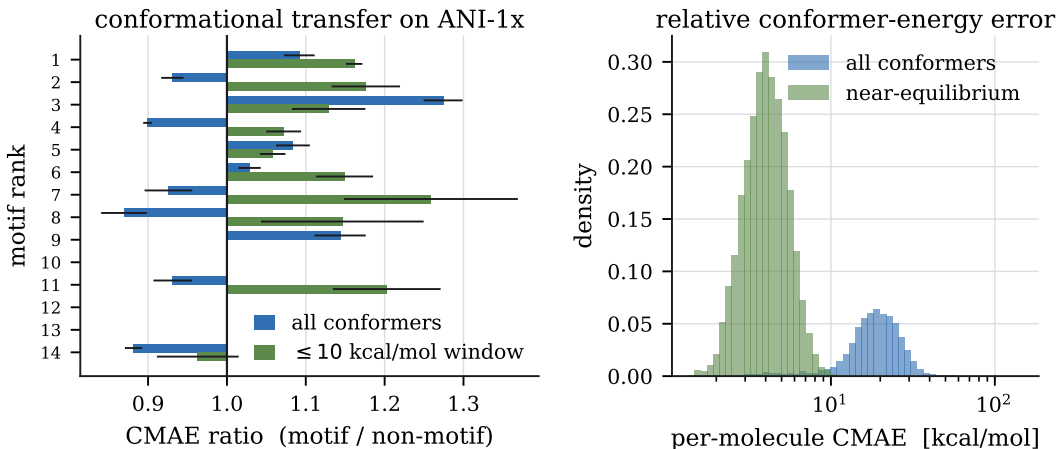


Figure 5: **Left:** per-motif ratio of conformational MAE between matching and non-matching ANI-1x molecules, over all sampled conformers and within the near-equilibrium window; bars start at unity and error bars are 95% intervals across seeds. **Right:** distribution of the per-molecule conformational MAE for the two strata.

remain *admissible* on the new models; they are displaced from the greedy selection, not disqualified, and the 4 environments that enter in their place (an *N*-substituted pyridine-type nitrogen, a tertiary amine, and two further strained bicyclics) are in the same chemical families. The turnover is between near-duplicate environments competing for the same coverage, not a change in what chemistry the library points at.

The calibration numbers move only slightly, and not all in the same direction: the single-model signals improve, in step with the accuracy — *R* alone gives $\text{ECE } 0.0387 \pm 0.0075$ against 0.0406 ± 0.0105 on the default models and HiSPEC gives 0.0185 ± 0.0017 against 0.0281 ± 0.0050 — while the five-model ensemble moves the other way, 0.0176 ± 0.0027 against 0.0125 ± 0.0007 . We have no clean explanation for that reversal; with 5 seeds the ensemble ECE is a noisier statistic than the single-model ones, and we report it as observed rather than curate it away. What does not change is the qualitative ordering: the ensemble still ranks errors substantially better than the single-model hierarchy (0.548 ± 0.005 against 0.369 ± 0.008 in rank correlation), so a better-initialised model does not make its own hierarchy a competitive substitute for an ensemble.

We draw two things from this. The failure structure reported in this paper is a property of the HIP-NN hierarchy under heavy-atom extrapolation and not of one initialisation choice; and a model that is 7% more accurate in distribution fails on the same chemistry, which is what one would expect if the motifs mark places where the two-interaction-layer local expansion is the wrong functional form rather than places where optimisation happened to fall short.

6 Where the analysis fails

Two kinds of failure deserve separate treatment: molecules the motif library does not explain, and molecules the uncertainty model gets wrong.

Failures outside the library. On the evaluation half the library matches 67.6% of molecules and 90.1% of failures, leaving 1,314 unexplained failures per seed. These are not a hidden second population: their mean absolute error is 1.276 kcal/mol against 2.948 kcal/mol for the failures the library does match, so what the library misses is the mild end of the tail rather than a distinct severe mode. Structurally they are the acyclic, heteroatom-poor remainder of QM9 — molecules with no ring system, few heteroatoms and little unsaturation, which fail for reasons the graph predicates in our candidate family cannot express. The obvious suspects are conformational: a specific dihedral, an intramolecular hydrogen bond, a close non-bonded contact. Our candidate vocabulary is deliberately graph-based so that motifs transfer to datasets where geometries differ, and this is the price.

Where HiSPEC is wrong. Figure 6(left) shows realised against predicted error for one seed. The estimator is a conditional-mean predictor, so the vertical scatter at fixed $\hat{u}$ is expected and irreducible;

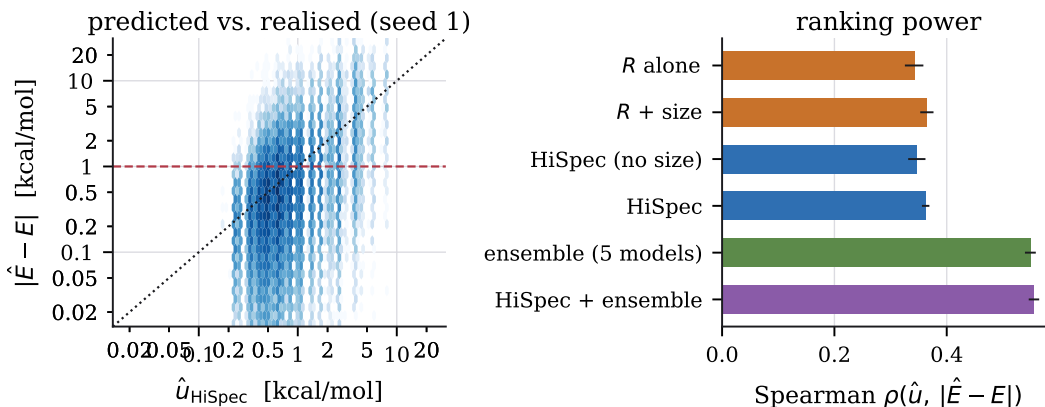


Figure 6: **Left:** realised against predicted error on the evaluation half for one seed; the dotted line is the identity, the dashed line the 1 kcal/mol threshold. **Right:** Spearman correlation between each uncertainty signal and the realised error, with 95% intervals across seeds.

what matters is the two tails. Of the molecules in the top uncertainty percentile, 4.7% turn out to have errors below 0.25 kcal/mol — false alarms, which cost a wasted reference calculation. More interesting is the opposite corner: of the molecules with errors above 2 kcal/mol, 12.1% receive an uncertainty below the median. These are the cases the hierarchy is blind to, and the reason is visible in the mechanism: the diagnostic measures whether the expansion *converges*, not whether it converges to the right answer. A molecule whose $\hat{E}^2$ is small relative to $\hat{E}^1$ scores as perfectly hierarchical even if $\hat{E}^1$ is itself wrong, and nothing in Eq. (2) can detect a well-behaved expansion around a badly-placed zeroth order.

That failure mode is exactly where the motif library earns its keep as a separate object rather than as another feature of the uncertainty model: 93.2% of these missed high-error molecules match at least one motif, against 67.6% of the evaluation set overall. A structural predicate can flag chemistry the model is silently confident about, which is the strongest practical argument we can make for keeping the two diagnostics distinct. It is also why we did not fold the motif indicators into HiSPEC: they would have improved the numbers, but the improvement would come from a supervised selection step and would no longer be a statement about what the hierarchy knows about itself.

7 Computational cost

All training was done on a single NVIDIA H200 per run. A heavy-atom-extrapolation seed takes 9.6 ± 0.1 s per epoch and 2.45 ± 0.15 h in total, running for 916 ± 58 epochs with the best validation score at epoch 815 ± 58 ; the paper reports about one minute per epoch on a P100 for the full QM9 set at the same architecture. The 3 50,000-molecule anchors ran for 1055 ± 141 epochs (7.20 ± 0.99 h each) and the 2 110,426 anchors for 1070 ± 130 epochs (16.38 ± 2.18 h each). The whole campaign is 20 runs and 109.8 GPU-hours.

The 110,426 anchors are the one place where we depart from the paper’s schedule: at ≈ 54 s/epoch a full $t_{\text{max}} = 2000$ run costs about 30 h, and we capped them at 1200 epochs. 1 of 2 runs reached that ceiling: one seed stopped early on the paper’s patience rule and the other ran to the cap. All other runs, at every smaller size, use the paper’s full $t_{\text{max}} = 2000$ and every one of them terminated on the patience rule before reaching it, so the reduced ceiling affects at most the one capped 110,426 seed and none of the analysis models; its consequences for the anchor comparison are discussed in §5.1.

Inference is cheap and this is the point of the estimator. Producing the entire hierarchy spectrum for a molecule — one forward pass, the per-atom per-order energies, R , and all 27 HiSPEC features — costs 0.06 ms per molecule at batch size 256, so the full 109,024-molecule extrapolation set takes 0.1 min per seed; the ANI-1x conformer sweep runs at 0.069 ms per conformer at batch size 1024. Relative to a bare energy prediction the overhead is a handful of matrix–vector products and some reductions: the read-out weights are already applied to compute the energy, and the extra cost is recombining terms the network has produced anyway. Evaluating the gradient-boosted regressor

on 27 features is negligible beside the network forward pass. The five-model ensemble baseline, by contrast, costs five times the training and five times the inference, which is the trade the last two rows of Table 6 quantify. Motif matching is a cheminformatics operation independent of the model: applying all 14 predicates to a molecule is a handful of RDKit substructure searches, milliseconds per molecule, and needs no forward pass at all.

8 Related work

Uncertainty quantification for machine-learned interatomic potentials has converged on a small number of families, and it is worth being explicit about where the present estimator sits relative to each.

Ensembles and query-by-committee. The dominant approach trains M independently initialised models and reads the spread of their predictions as an uncertainty [Lakshminarayanan et al., 2017, Musil et al., 2019]. In atomistic chemistry it is the engine of essentially all active-learning pipelines, including the ANI series [Smith et al., 2018], moment-tensor potentials [Podryabinkin and Shapeev, 2017] and DP-GEN [Zhang et al., 2020]. It is accurate, requires no architectural change, and costs M times as much at both training and inference; recent work reduces that cost by sharing a trunk across ensemble heads [Kellner and Ceriotti, 2024]. We use a five-model ensemble purely as a reference point. The comparison we care about is not whether a single model can beat an ensemble — it does not — but how much of the ensemble’s ranking power is already latent in one model’s own decomposition, and at what fraction of the cost.

Bayesian and kernel methods. Gaussian-process potentials [Bartók et al., 2010, Deringer et al., 2021] come with a posterior variance, which Vandermause et al. [2020] exploit for on-the-fly active learning; Monte-Carlo dropout [Gal and Ghahramani, 2016] gives a cheap Bayesian approximation for neural potentials [Wen and Tadmor, 2020]. These estimate uncertainty by sampling a distribution over functions; HiSPEC samples nothing and is deterministic given the model.

Single-model deterministic estimators. Mean–variance networks [Nix and Weigend, 1994], evidential regression [Amini et al., 2020, Soleimany et al., 2021] and latent-space distance metrics [Janet et al., 2019] obtain uncertainty from one model, either by adding a head that predicts a variance or by measuring how far a query is from the training data in feature space. Zhu et al. [2023] obtain fast estimates from gradients of the network output. HiSPEC is in this family in cost but differs in kind: it adds no head, changes no loss, and is not a distance to the training set. Its inputs are the terms of a physically motivated many-body expansion that the architecture already computes and already regularises, so the signal is a statement about the model’s internal self-consistency rather than about the geometry of its feature space. The closest published relative is the original observation of Lubbers et al. [2018] that R correlates with error, which is precisely the baseline we improve on; the improvement comes from undoing three specific aggregations in Eq. (2) rather than from adding an external estimator.

Conformal prediction and recalibration. Conformal methods [Hu et al., 2022] wrap any point predictor in finite-sample-valid intervals given an exchangeable calibration set; isotonic and Platt-style recalibration [Zadrozny and Elkan, 2002, Guo et al., 2017, Kuleshov et al., 2018] rescale an existing score to match observed frequencies. Our isotonic step belongs to the second family and is used in the standard way: to move a score onto the error scale of a shifted distribution. We make no exchangeability claim and therefore no coverage guarantee — our calibration statements are empirical, on a held-out half of the extrapolation set. A conformal wrapper on top of HiSPEC would be a natural extension and is not attempted here.

Evaluating uncertainty for molecules. The methodology of comparing uncertainty estimators for molecular properties has itself been studied [Tran et al., 2020, Scalia et al., 2020, Hirschfeld et al., 2020]. We follow the recommendations of that literature in reporting a reliability diagram over equal-count bins, a scalar calibration error, and a ranking metric that is invariant to monotone rescaling, rather than any one of them alone.

Interpretability and failure analysis of potentials. Attribution of model behaviour to chemistry has mostly taken the form of visualising learned per-atom quantities: local chemical potentials in DTNN [Schütt et al., 2017], learned partial charges in HIP-NN [Sifain et al., 2018] and AIMNet [Zubatyuk et al., 2019]. Those studies read chemistry *out of* a trained model. We do something narrower and more testable: we ask which chemical predicates *predict* a specific internal diagnostic and a specific failure, register the candidate predicates before looking, control the false-discovery rate across them, and validate on a disjoint half. To our knowledge the resulting object — a statistically validated, compact library of structural motifs tied to a named architectural diagnostic, with a per-motif conformational transfer rate — has no direct precedent in the potential-energy-surface literature, though the ingredients (SMARTS catalogues, circular fingerprints [Rogers and Hahn, 2010], Benjamini–Hochberg control [Benjamini and Hochberg, 1995], post-selection discipline [Berk et al., 2013, Taylor and Tibshirani, 2015]) are all standard in their own fields.

Newer architectures. Equivariant potentials [Batzner et al., 2022, Batatia et al., 2022, Musaelian et al., 2023] now reach lower QM9 and MD17 errors than HIP-NN, and HIP-NN itself has been extended with tensor sensitivities [Chigaev et al., 2023]. We deliberately hold the 2018 architecture fixed: the object of study is what the hierarchy reveals about itself, and changing the architecture would change the decomposition whose failure we are characterising. Whether an analogous diagnostic exists for the body-ordered expansions of MACE-like models is an open question that this work is intended to motivate.

9 Limitations and conclusions

What was held fixed, and what that costs. The HIP-NN architecture and optimisation protocol are those of Lubbers et al. [2018] throughout, so nothing here speaks to whether a different architecture would have different failure modes. Two implementation details differ from the 2018 description; both are measured rather than merely disclosed. The initialisation of the higher-order read-out weights follows the authors’ current `hippynn` package, which omits the σ_E factor of the paper’s Sec. II.6 — a difference worth about 7% in MAE (§5.2) that leaves the failure structure unchanged (§5.9). The two 110,426-molecule anchor runs were given a reduced epoch ceiling for compute reasons (§7); one of them reached that ceiling, so we cannot claim it is fully converged, while every other run in the campaign — including all the models the analysis uses — stopped earlier on the paper’s own patience rule. Neither difference touches the motif or calibration analysis, which is conducted on ≤ 8 -heavy-atom models trained to the paper’s full $t_{\max} = 2000$ under both initialisations.

Limitations. Four are worth stating plainly. *First*, the extrapolation studied here is a size extrapolation within a single dataset and a single level of theory; larger shifts — different elements, charged species, condensed phases — may produce failure modes the library does not contain. *Second*, the motif predicates are graph predicates. They cannot express conformational features such as a specific dihedral or a close contact, which is precisely why the ANI-1x results in §5.8 split the way they do; a geometry-aware motif vocabulary is the obvious next step. *Third*, the calibration claims are empirical, on a held-out half of the extrapolation set, and carry no finite-sample coverage guarantee; a conformal wrapper would supply one at the cost of an exchangeability assumption that is exactly what heavy-atom extrapolation violates. *Fourth*, the transfer datasets are computed at different levels of theory from the training data, so the conformational errors of §5.8 contain a functional-mismatch component that our within-molecule centring removes only to first order.

Conclusions. HIP-NN’s hierarchical decomposition was introduced as a route to accuracy and, secondarily, as a source of a self-diagnostic. The self-diagnostic turns out to be considerably richer than the single scalar that has been read off it. Undoing the three aggregations hidden in Eq. (2) — over atoms, over orders, and over energy scale — yields an uncertainty estimate that is well calibrated on chemistry the model was not trained on, from one forward pass of one model; and, for the motifs whose atoms can be localised, it points at the specific atoms carrying the failure often enough to make a chemical explanation possible — though §5.5 shows that particular reading is the least seed-stable thing we report. The resulting motif library is compact enough to be read, statistically supported on a disjoint half of the test set, and partially transferable to off-equilibrium geometries. The general lesson we would draw is that a physically structured decomposition is worth more than the accuracy

it buys: an architecture that writes its answer as a sum of interpretable terms can be interrogated about which term went wrong, and that question has a chemical answer.

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A Reproducing the numbers

All code is in `proposal/code/` and all machine-readable results in `proposal/logs/`. The pipeline is

1. `qm9_prepare.py` — parse the `dsgdb9nsd` release, apply the two pruning lists, form atomisation energies, write `data_cache/qm9_clean.npz` and `logs/qm9_prepare.json`.
2. `atom_mapping.py` — align xyz atom order to RDKit graph order (`logs/atom_mapping.json`).
3. `train_hipnn.py` — one HIP-NN run; `launch_training.sh` and `launch_extra.sh` launch the campaign. Each run writes `meta.json` (hyperparameters, seeds, parameter counts, host, GPU, timestamps), `partition.json` (split sizes, heavy-atom ranges, pairwise QM9-index overlaps), `split_indices.npz`, `training_log.txt`, `epoch_metrics.json` (per-epoch train and validation metrics), `result.json` and the checkpoints `best_model.pt` (0.9 MB) and `best_checkpoint.pt`.
4. `evaluate_model.py` — per-molecule $\hat{E}$, E , R , $\{r_i\}$ and $\{\hat{E}_i^n\}$ for a split, with the reconstruction self-check written to `*_check.json`.
5. `ani1x_prepare.py`, `rmd17_prepare.py`, `evaluate_conformers.py` — the transfer sets and their predictions.
6. `run_analysis.py`, `ani1x_transfer.py`, `rmd17_transfer.py`, `collect_runs.py` — every statistic in the paper.
7. `figures.py`, `make_tables.py` — figures, table bodies and `tex/macros.tex`. Every number in the manuscript is a macro expanded from that file, so the text cannot drift from the logs.

B The HiSPEC feature vector

Table 8: The HiSPEC feature vector, all 27 components (the last five rows contribute two features each). Every entry is a deterministic function of one forward pass of one model and requires no reference energy.

feature	what it restores
$\log_{10} R$	Eq. (2) here, Eq. (18) of Lubbers et al. [2018] : their signal
R/N_{atom}	intensive form, removes the size trend
$\max_i r_i$	worst single atom (undoes the atom sum)
$q_{90}(r_i)$	90th percentile of the per-atom terms
$\text{sd}_i(r_i)$	spread of the per-atom terms
$\langle \mathbf{1}[r_i > \frac{1}{2}] \rangle_i$	fraction of atoms with a non-decaying expansion
$R^{(1)}/N_{\text{atom}}$	order-1 part of R (undoes the order sum)
$R^{(2)}/N_{\text{atom}}$	order-2 part of R
$\log_{10} \sum_i \hat{E}_i^1 $	absolute size of the order-1 correction
$\log_{10} \sum_i \hat{E}_i^2 $	absolute size of the order-2 correction
$\log_{10} \max_i \hat{E}_i^1 $	largest single order-1 contribution
$\log_{10} \max_i \hat{E}_i^2 $	largest single order-2 contribution
$\log_{10} (\sum_i (\hat{E}_i^2)^2 / \sum_i (\hat{E}_i^1)^2)$	molecular decay ratio (undoes the scale invariance)
$\log_{10} (\sum_i \hat{E}_i^1 / \sum_i \hat{E}_i^0)$	correction size relative to the dressed-atom term
molecular hierarchicality	hippynn’s ratio-of-sums variant of R
N_{atom}	molecule size
N_{heavy}	heavy-atom count
$\langle r_i \rangle_H, N_H$	mean per-atom term over the H atoms, and their count
$\langle r_i \rangle_C, N_C$	mean per-atom term over the C atoms, and their count
$\langle r_i \rangle_N, N_N$	mean per-atom term over the N atoms, and their count
$\langle r_i \rangle_O, N_O$	mean per-atom term over the O atoms, and their count
$\langle r_i \rangle_F, N_F$	mean per-atom term over the F atoms, and their count

C Per-seed results

Table 9: Per-seed results for the heavy-atom extrapolation protocol. MAE in kcal/mol; ECE and precision/recall are for HiSPEC on the evaluation half.

run	seed	epochs	best ep.	valid	in-dist.	9-heavy	ECE	recall
heavy8_s1	1	1064	963	0.3453	0.3562	0.860	0.0278	0.591
heavy8_s2	2	773	672	0.3891	0.4096	0.893	0.0232	0.608
heavy8_s3	3	1040	939	0.3210	0.3508	0.809	0.0183	0.582
heavy8_s4	4	817	716	0.3633	0.3487	0.813	0.0240	0.639
heavy8_s5	5	887	786	0.3603	0.3915	0.891	0.0471	0.647

D Supplementary conformational transfer: rMD17

ANI-1x supplies the sample size the per-motif transfer test needs, but every one of its molecules is chemistry HIP-NN has not been trained on, so its conformational errors mix a geometry effect with a chemistry effect. rMD17 [[Christensen and von Lilienfeld, 2020](#)] supplies the complementary control: five of its ten molecules (benzene, ethanol, malonaldehyde, toluene, uracil) have at most eight heavy atoms and therefore lie inside the training range by size, so any failure on their MD-sampled conformers is attributable to geometry alone. We draw 50,000 conformations in total (5,000 per molecule, uniformly at random with seed 20240617) and score them with Eq. (4).

The conformational error is large for every molecule (9.26 ± 0.38 kcal/mol on average), and larger for the five molecules outside the training size range (11.29 kcal/mol) than for the five inside it (7.23 kcal/mol) — a model trained only on minimised geometries does not describe a room-temperature potential-energy surface, whatever its in-distribution MAE. What matters here is the second column of Table 10: within a single molecule, where chemistry is held exactly fixed and

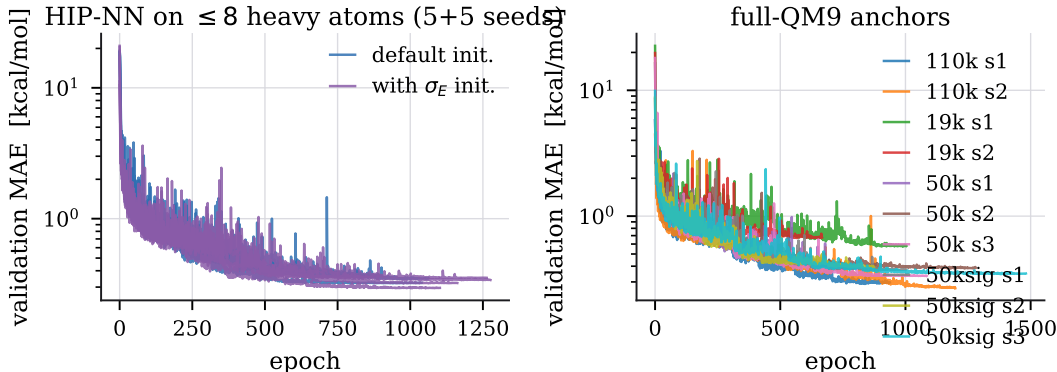


Figure 7: Validation MAE against epoch. Left: the 5 seeds of the heavy-atom extrapolation protocol under hippynn’s default initialisation, and the 5 run with the paper’s σ_E factor restored (§5.2). Right: the full-QM9 anchors run under the paper’s own protocol, including the σ_E variants at 50,000. The visible steps are the plateau-triggered halvings of the learning rate.

Table 10: rMD17 supplementary transfer. “In size” marks molecules with at most eight heavy atoms, i.e. inside HIP-NN’s training range. CMAE is the within-molecule relative conformer-energy error of Eq. (4); ρ are within-molecule Spearman correlations between the uncertainty signal and the per-conformer error. Means over 5 seeds.

molecule	heavy	in size	motifs matched	CMAE	$\rho(R)$	$\rho(\text{HiSPEC})$
ethanol	3	yes	none	4.58	0.086	0.073
malonaldehyde	5	yes	none	5.76	0.047	0.049
benzene	6	yes	4, 6	5.43	0.077	0.093
toluene	7	yes	4, 6	8.48	0.071	-0.033
uracil	8	yes	2, 4, 6	11.91	0.116	0.056
naphthalene	10	no	1, 4, 6	10.65	0.069	0.077
salicylic	10	no	4, 6	10.68	0.078	0.052
paracetamol	11	no	4, 6	11.12	0.084	0.061
aspirin	13	no	2, 4, 6	11.76	0.045	0.062
azobenzene	14	no	1, 4, 6	12.26	-0.015	-0.017

only the geometry varies, HiSPEC still ranks conformers by error with a mean Spearman correlation of 0.047 ± 0.006 against 0.066 ± 0.007 for R alone, and 0.047 within the in-training-size subset. The hierarchy-failure signal is therefore not purely a chemical-composition detector: it responds to geometry at fixed composition. With ten molecules the association between the number of motifs a molecule matches and its conformational error (Spearman 0.79, $p = 0.01$) is reported descriptively only.

E The full candidate family

The complete scored candidate family — all 873 predicates with their discovery-half coverage, risk ratio, attribution, effect size and Benjamini–Hochberg q -values, and a flag for whether each was admissible — is in `logs/motif_candidates_all.json.gz`. The library reported in Table 3 is the greedy selection from the 218 admissible members; no admissible candidate was removed by hand.